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Assignment # 2

Subject: Database Systems

Question # 1(A)

❖ The set of functional dependencies for a relation $R(A,B,C,D,E,F,G)$ is:
 $F = \{AD \rightarrow BF, CD \rightarrow EGC, BD \rightarrow F, E \rightarrow D, F \rightarrow C, D \rightarrow F\}$

I find all candidate keys.

ii. Find F-closure.

Answer # 1(A)

1. F-closure

Firstly, we find F-closure, which is given below:

1. $AD^+ = ABCDEFG$
2. $BD^+ = BCDEFG$
3. $CD^+ = CDEG$
4. $D^+ = CDEFG$
5. $E^+ = CDEFG$
6. $F^+ = CF$

2. Candidate keys

From the above closure, we get that AD is the only minimal key $\Rightarrow AD^+ = ABCDEFG$.

Question # 1(B)

❖ The set of functional dependencies for a relation $R(A,B,C,D,E,F,G,H,I,J)$ is:
 $F = \{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$

What is key for R?

ii. Decompose R into 2NF and then into 3NF.

Answer # 1(B)

1. Relational Key

Firstly, we find X-closure, which is given below:

1. $AB^+ = ABCDEFGHIJ$
2. $A^+ = ADEIJ$
3. $B^+ = BFGH$
4. $D^+ = DIJ$
5. $F^+ = FGH$

Now, from the above closure, we get that AB is the only minimal key $\Rightarrow AB^+ = ABCDEFGHIJ$

2. Decomposition

1NF

Since there are no multiple repeating attributes in the given relation thus the given relation is already in 1NF.

2NF

1. Since AB is the candidate key, it is:

$R1 (\underline{A}, B, C), AB \rightarrow C$

$R2 (\underline{A}, D, E), A \rightarrow DE$

$R3 (\underline{B}, F), B \rightarrow F$

2. Since $B \Rightarrow F$, thus,

$R4 (\underline{E}, G, H), F \rightarrow GH$

3. Since $A \Rightarrow DE$, and $A \Rightarrow D$, thus,

$R5 (\underline{D}, I, J), D \rightarrow IJ$

Now, from the above observation, we get the 2NF Form of the given relation, which is given below:

1) $R1 (\underline{A}, B, C)$

2) $R2 (\underline{A}, D, E)$

3) $R3 (\underline{B}, F)$

4) $R4 (\underline{E}, G, H)$

5) $R5 (\underline{D}, I, J)$

3NF

Since there are no transitive dependencies in the above 2NF relation thus the 3NF form of the relation would be the same as the 2NF form of the relation

Question # 2

The set of functional dependencies for relation $R(A,B,C,D,E)$ is:

$F = \{AB \rightarrow C, AB \rightarrow D, D \rightarrow A, BC \rightarrow D, BC \rightarrow E\}$

a. Find all candidate keys.

b. Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF).

c. Is this relation in BCNF? If not, show all dependencies that violate it.

d. Is this relation in 3NF? If not, show all dependencies that violate it.

Answer # 2(A)

1. Candidate keys

Firstly, we find X -closure, which is given below:

1. $AB^+ = ABCDE$

2. $BC^+ = ABCDE$

3. $D^+ = AD$

From the above closure, we get the following candidate keys:

1) $AB^+ = ABCDE$

2) $BC^+ = ABCDE$

Answer # 2(B)

1. Best Normalises Form

The best normalised form of the given relation is 3NF.

2. Reason

The only non-trivial FD with a non-key determinant is $D \rightarrow A$. Since A is a prime attribute, $D \rightarrow A$ does not violate the 3NF condition. Therefore, the best normal form that satisfies the relation R is in 3NF.

Answer # 2(C)

1. BCNF check

No. It is **not** in BCNF.

2. Reason

The dependencies that violate the BCNF conditions are $D \rightarrow A$.

Answer # 2(D)

1. 3NF check

Yes, the given relation is in 3NF form.

Question # 3

- 14.14 Examine the Patient Medication Form for the Wellmeadows Hospital case study (see Appendix B) shown in Figure 14.18.
- Identify the functional dependencies represented by the attributes shown in the form in Figure 14.18. State any assumptions that you make about the data and the attributes shown in this form.
 - Describe and illustrate the process of normalizing the attributes shown in Figure 14.18 to produce a set of well-designed 3NF relations.
 - Identify the primary, alternate, and foreign keys in your 3NF relations.

Wellmeadows Hospital Patient Medication Form							
Patient Number: <u>P10034</u>							
Full Name: <u>Robert MacDonald</u>		Ward Number: <u>Ward 11</u>					
Bed Number: <u>84</u>		Ward Name: <u>Orthopaedic</u>					
Drug Number	Name	Description	Dosage	Method of Admin	Units per Day	Start Date	Finish Date
10223	Morphine	Pain Killer	10mg/ml	Oral	50	24/03/13	24/04/14
10334	Tetracycline	Antibiotic	0.5mg/ml	IV	10	24/03/13	17/04/13
10223	Morphine	Pain Killer	10mg/ml	Oral	10	25/04/14	02/05/15

Answer # 3(A)

1. Functional Dependencies

From the given data, the functional dependencies that can be formed are as follows:

1. PatientNumber → FullName, BedNumber, WardNumber, WardName
2. WardNumber → WardName
3. DrugNumber → DrugName, Description, Dosage, Method of Administration
4. (PatientNumber, DrugNumber, StartDate) → UnitsPerDay, FinishDate

2. Assumptions

The assumptions that can be made from the given tables are as follows:

1. Each drug will have these details: “name”, “description”, “dosage”, and “method”.
2. A patient can be prescribed the same drug more than once.
3. Every ward has a unique ward number assigned to it.
4. Every patient had a unique patient number assigned to them.
5. Each patient is assigned to only one ward at a time.
6. Each ward has multiple beds and thus has multiple patients.

Answer # 3(B)

1. Description/Illustration

The process of normalising the attributes shown in Figure 14.18 to produce a set of well-designed 3NF relations consists of the steps, which are as follows:

UNF

The original form mixes patient, ward, and drug data in a **repeating group**, which is as follows:

1. (PatientNumber, FullName, BedNumber, WardNumber, WardName, DrugNumber, Name, Description, Dosage, Method, UnitsPerDay, StartDate, FinishDate)

1. Step #1

1NF

The first step consists of removing the repeating groups by creating a separate row for each drug entry (decomposing the relation)

1. PATIENT (PatientNumber, FullName, BedNumber, WardNumber, WardName)
2. MEDICATION (PatientNumber, DrugNumber, DrugName, Description, Dosage, Method, UnitsPerDay, StartDate, FinishDate)

2. Step #2

2NF

The second step consists of removing partial dependencies from the relation by creating a separate row for each drug entry, these occur when non-key attributes depend on part of a composite key.

1. PATIENT (PatientNumber, FullName, BedNumber, WardNumber, WardName)
2. Ward(WardNumber, WardName)
3. Drug(DrugNumber, DrugName, Description, Dosage, Method)
4. Prescription(PatientNumber, DrugNumber, StartDate, UnitsPerDay, FinishDate)

3. Step #3

3NF

The third step involves removing transitive dependencies from the relations by creating separate rows.

After reviewing the relations created in step 2, we discovered the following:

1. All non-key attributes are fully functionally dependent on the whole primary key
2. There are no transitive dependencies

2. 3NF Normalisation

By following the definitions of Normalization Forms like (1NF, 2NF, and 3NF) the above unnormalized relation can be converted into 3NF form, which are as follows:

1. Patient(PatientNumber, FullName, BedNumber, WardNumber)
2. Ward(WardNumber, WardName)
3. Drug(DrugNumber, DrugName, Description, Dosage, MethodOfAdmin)
4. Prescription(PatientNumber, DrugNumber, StartDate, UnitsPerDay, FinishDate)

Answer # 3(C)

1. Patient

1. **Primary Key:** PatientNumber
2. **Alternate Key:** (WardNumber, BedNumber)
(Assuming each bed number is unique within a ward)
3. **Foreign Key:** WardNumber → Ward(WardNumber)

2. Ward

1. **Primary Key:** WardNumber
2. **Alternate Key:** None
3. **Foreign Key:** None

3. Drug

1. **Primary Key:** DrugNumber

2. **Alternate Key:** Possibly DrugName (if guaranteed unique)
3. **Foreign Key:** None

4. Prescription

1. **Primary Key:** (PatientNumber, DrugNumber, StartDate)
2. **Alternate Key:** None
3. **Foreign Keys:** PatientNumber, DrugNumber

Question # 4

Question#4 [10]

A leasing company lease flats to its clients. The sample data of the company is given in the table below. A place number (placeNo) uniquely identifies each single room in all flats and is used when leasing a room to a student. Perform the following tasks:

- Identify the functional dependencies that exist between the columns of the table and identify the primary key and any alternate keys (if present) for the table.
- Describe why the table is not in 3NF.
- The table is susceptible to update anomalies. Provide examples of how insertion, deletion, and modification anomalies could occur on this table.

leaseNo	bannerID	placeNo	fName	lName	startDate	finishDate	flatNo	flatAddress
10003	B017706	78	Jane	Watt	01/09/2010	30/06/2011	F56	34 High Street, Paisley
10259	B017706	88	Jane	Watt	01/09/2011	30/06/2012	F78	111 Storrie Road, Paisley
10364	B013399	89	Tom	Jones	01/09/2011	30/06/2012	F78	111 Storrie Road, Paisley
10566	B012124	102	Karen	Black	01/09/2011	30/06/2012	F79	120 Lady Lane, Paisley
11067	B034511	88	Steven	Smith	01/09/2012	30/06/2013	F78	111 Storrie Road, Paisley
11169	B013399	78	Tom	Jones	01/09/2012	30/06/2013	F56	34 High Street, Paisley

Answer # 4(A)

1. Functional Dependencies

The functional dependencies between the columns of the table are given below:

1. leaseNo → bannerID, placeNo, fName, lName, startDate, finishDate, flatNo, flatAddress
2. placeNo → flatNo, flatAddress
3. flatNo → flatAddress (transitive dependency via placeNo)
4. bannerID → fName, lName

2. Primary Key

The Primary key in the given table is leaseNo.

3. Alternate Key

The alternate key in the given table is a composite key, which is (placeNo, startDate); this is only true when a room cannot be leased twice on the same start date.

Answer # 4(B)

1. 3NF Check

The table violates 3NF due to transitive dependencies, which are as follows:

1. Non-key attributes (fName, lName) depend on bannerID, which is not a candidate key.
 - $\text{leaseNo} \rightarrow \text{bannerID} \rightarrow \text{fName, lName}$
2. Non-key attributes (flatNo, flatAddress) depend on placeNo, which is not a candidate key.
 - $\text{leaseNo} \rightarrow \text{bannerID} \rightarrow \text{fName, lName}$

Answer # 4(C)

1. Update Anomalies

The update anomalies that can be formed in this table are as follows:

1. Insertion Anomaly

The insurability anomalies that can occur in the table are as follows:

1. Cannot add a new flat without a corresponding lease.
2. Cannot add a student (bannerID) unless they have a leaseNo.

2. Deletion Anomaly

The deletion anomalies that can occur in the table are as follows:

1. Deleting lease 10566 (Karen Black) would remove all information about flat F79

3. Modification Anomaly

The Modification anomalies that can occur in the table are as follows:

1. Changing the address of flat F78 requires updating all 3 rows where F78 appears. If missed, data becomes inconsistent.

Question # 5

4-54. Explore the data included in Table 4-9.

Assume that the primary key of this relation consists of two components: Author's ID (AID) and Book number (BNbr). The relation includes data regarding authors, books, and publishers. In addition, it tells what an individual author's per book royalty amount is in the case of multi-authored books.

Your task is to:

- a. Identify the functional dependencies between the attributes.
- b. Identify the normal form in which the relation currently is.

TABLE 4-9 Author Book Royalties

<u>AID</u>	<u>ALname</u>	<u>AFname</u>	<u>Alnst</u>	<u>BNbr</u>	<u>BName</u>	<u>BPublish</u>	<u>PubCity</u>	<u>BPrice</u>	<u>AuthBRoyalty</u>
10	Gold	Josh	Sleepy Hollow U	106	JavaScript and HTML5	Wall & Vintage	Chicago, IL	\$62.75	\$6.28
				102	Quick Mobile Apps	Gray Brothers	Boston, MA	\$49.95	\$2.50
24	Shippen	Mary	Green Lawns U	104	Innovative Data Management	Smith and Sons	Dallas, TX	\$158.65	\$15.87
				106	JavaScript and HTML5	Wall & Vintage	Indianapolis, IN	\$62.75	\$6.00
32	Oswan	Jan	Middlestate College	126	Networks and Data Centers	Grey Brothers	Boston, NH	\$250.00	\$12.50
				180	Server Infrastructure	Gray Brothers	Boston, MA	\$122.85	\$12.30
				102	Quick Mobile Apps	Gray Brothers	Boston, MA	\$45.00	\$2.25

Answer # 5(A)

1. Functional Dependencies

The functional dependencies in the given table are as follows:

1. $AID \rightarrow ALname, AFname, Alnst$
2. $BNbr \rightarrow BName, BPublish, PubCity, BPrice$
3. $BPublish \rightarrow PubCity$
4. $(AID, BNbr) \rightarrow AuthBRoyalty$

Answer # 5(B)

1. Normalised Form

The relation is currently in 1NF form.

2. Reason

All attributes are atomic, and there are no repeating groups.

Question # 6

4-58. A bus company is responsible for offering public transportation in the suburbs of a large metropolitan area. The company has significant data management requirements: It needs to keep track of its 150 vehicles, 400 drivers, 60 bus routes, and hundreds of scheduled departures every day. In addition, it is essential for the company to know which drivers are certified to drive which buses. The data that the company has available include the following attributes:

RouteID, RouteStartPoint, RouteEndPoint,
RouteStandardDrivingTime, ScheduleDate,
ScheduledDepTime, ScheduledArrTime,
DriverID, DriverFName, DriverLName,
DateDriverJoinedCompany, DriverDOB, VehicleID,
VehicleMake, VehicleModel, VehiclePassengerCapacity,
DriverCertStartDate, DriverCertEndDate.

Sample data for this set of attributes are as follows:

28, Grand Avenue, Madison Street, 38, {9/12/2015,
8.30, 9.18, 8273, Mary, Smith, 5/2/2007, 3/23/1974,
1123, GreatTrucks, CityCoach, 58, 6/10/2015,
6/9/2016 | 9/12/2015, 9.30, 10.12, 7234, John, Jones,
10/12/2011, 12/15/1991, 5673, GreatTrucks, CityCoach
2, 62, 4/12/2015, 4/11/2016 | 9/12/2015, 10.30, 11.08,
2343, Pat, Moore, 2/24/1982, 1/19/1958, 4323,
PowerTransport, MidiBus, 32, 8/20/2015, 8/19/2016}

Note that the information for specific bus schedules (starting with the attribute ScheduleDate) is repeated three times in the sample data set and is separated by the "|" symbol. Also, take into account that in this case, the certification is specific to a particular vehicle driver pair.

Based on the facts stated above,

- Identify the functional dependencies between the attributes.
- Identify the reasons why this set of data is not in 3NF and indicate the normal form (if any) it is in.

- Including all intermediate stages, organize the attributes into a set of 3NF relations.
- Draw an ER diagram based on the normalized relations.
- Based on the ER diagram you just drew and the case narrative, explore the areas in which there could be opportunities to expand the data model to achieve better tracking of the company's operations or improved clarity, such as maintaining more detailed route information.

Answer # 6(A)

1. Functional Dependencies

The functional dependencies in the given attributes are as follows:

1. $RouteID \rightarrow RouteStartPoint, RouteEndPoint, RouteStandardDrivingTime$
2. $DriverID \rightarrow DriverFName, DriverLName, DateDriverJoinedCompany, DriverDOB$
3. $VehicleID \rightarrow VehicleMake, VehicleModel, VehiclePassengerCapacity$
4. $(DriverID, VehicleID) \rightarrow DriverCertStartDate, DriverCertEndDate$
5. $(RouteID, ScheduleDate, ScheduleDepTime) \rightarrow ScheduleArrTime, DriverID, VehicleID$

Answer # 6(B)

1. Normalised Form

The given data is not in any normalisation form.

2. Reason

1. 1NF (First Normal Form):

1. Data has repeating groups (multiple schedules per route).

2. 2NF (Second Normal Form):

1. Even if made atomic, a table with a composite key (e.g., $(RouteID, ScheduleDate)$) would have **partial dependencies**, like: $RouteID \rightarrow RouteStartPoint$

3. 3NF (Third Normal Form):

1. Transitive dependencies: E.g., $RouteStartPoint \rightarrow RouteStandardDrivingTime$

Answer # 6(C)

1. Stages For 3NF

3 stages lead to the 3NF form, which are given below:

1. Stage #1

The first stage involves removing repeating groups and making all attributes atomic.

1NF

The 1NF form is as follows:

1. **ROUTE:** $(RouteID, RouteStartPoint, RouteEndPoint, RouteStandardDrivingTime)$
2. **SCHEDULE:** $(RouteID, ScheduleDate, ScheduleDepTime, ScheduleArrTime, DriverID, VehicleID, DriverFName, DriverLName, DateDriverJoinedCompany, DriverDOB, VehicleMake, VehicleModel, VehiclePassengerCapacity, DriverCertStartDate, DriverCertEndDate)$

2. Stage #2

The second stage involves removing partial dependencies, which occur when non-key attributes depend only on a part of a composite key.

2NF

The 2NF form is as follows:

1. *ROUTE: (RouteID, RouteStartPoint, RouteEndPoint, RouteStandardDrivingTime)*
2. *DRIVER: (DriverID, DriverFName, DriverLName, DateDriverJoinedCompany, DriverDOB)*
3. *VEHICLE: (VehicleID, VehicleMake, VehicleModel, VehiclePassengerCapacity)*
4. *CERTIFICATION: (DriverID, VehicleID, DriverCertStartDate, DriverCertEndDate)*
5. *SCHEDULE: (RouteID, ScheduleDate, ScheduleDepTime, ScheduleArrTime, DriverID, VehicleID)*

3. Stage #3

The third stage involves removing transitive dependencies where non-key attributes depend on other non-key attributes.

3NF

Since the 2NF has no Transitive dependencies, the 3NF will be the same as the 2NF.

Question # 7

Question# 7[15]:

The University Enrollment System tracks information about students, courses, instructors, departments, and enrollment details in a single table. This table is poorly designed and contains redundant and repeated data. Each time a student enrolls in a course, all related information about the student, instructor, course, and department is repeated in a new row. This leads to issues such as redundancy, anomalies (update, insertion, and deletion), and a lack of scalability.

Normalize the table given below by organizing the data into multiple related tables to eliminate redundancy, maintain data integrity, and scale efficiently for thousands of students, courses, and departments.

Attributes:

RollNo , Name, Email, Degree, CourseCode (CC) , CourseTitle (CT), InstructorID (IID), InstructorName (IN), Designation, Deptno (DN), DName, DHead, EnrollmentDate (EDate), Grade, Classroom (CR), Schedule

Sample Data

Roll No	Name	Degree	CC	Course Title	IID	Instructor Name	Instructor Email	DN	DName	DHead	EDate	Grade	CR	Schedule
1	Alice Smith	BS(SE)	101	Data Structures	501	Dr. John	john@mail.com	201	Software Engineering	Dr. Brown	01/15/2025	A	101	Mon/Wed 9:00-10:30
2	Bob Johnson	BS(CS)	102	Operating Systems	502	Dr. Susan	susan@mail.com	202	Computer Science	Dr. White	01/16/2025	B+	102	Tue/Thu 11:00-12:30
3	Nick Knight	BS(CS)	102	Operating Systems	502	Dr. Susan	susan@mail.com	202	Computer Science	Dr. White	01/16/2025	B	102	Tue/Thu 11:00-12:30

Answer # 7

1. Normalisation

1. Stage #1

The relation obtained from the table is as follows:

UNF

1. **UNIVERSITY_ENROLLMENT**(RollNo, Name, Email, Degree, CourseCode, CourseTitle, InstructorID, InstructorName, InstructorEmail, DeptNo, DName, DHead, EnrollmentDate, Grade, Classroom, Schedule)

2. Stage #2

The 1NF form of the relation obtained from the table is as follows:

1NF

1. STUDENT (RollNo, Name, Email, Degree)
2. ENROLLMENT (RollNo, CourseCode, CourseTitle, InstructorID, InstructorName, InstructorEmail, DeptNo, DName, DHead, EnrollmentDate, Grade, Classroom, Schedule)

3. Stage #3

The 2NF form of the relation obtained from the table is as follows:

2NF

1. STUDENT (RollNo, Name, Email, Degree)
2. COURSE (CourseCode, CourseTitle, InstructorID, Classroom, Schedule)
3. INSTRUCTOR (InstructorID, InstructorName, InstructorEmail, DeptNo)
4. DEPARTMENT (DeptNo, DName, DHead)
5. ENROLLMENT (RollNo, CourseCode, EnrollmentDate, Grade)

4. Stage #4

The 3NF form of the relation obtained from the table is as follows:

3NF

After reviewing the relations created in stage 3, we find that:

- All non-key attributes are fully functionally dependent on the primary key.
 - There are no transitive dependencies.
-

Question # 8

Question#8 [15]

VisionEstate is a property management company overseeing various residential properties across different cities in Pakistan. The company conducts routine property inspections to ensure the upkeep of these residences. When staff members need to carry out these assessments, they are assigned a corporate vehicle for use during the inspection day. However, a single vehicle may be designated to multiple staff members as needed throughout the workday. A staff member might assess numerous properties on a specific date, yet each property undergoes assessment only once on that particular day. Samples of the VisionEstate Property Evaluation Report are depicted in table below. IDate and ITime stands for InspectionDate and InspectionTime respectively.

Sample Data

<u>PropertyNo</u>	<u>PAddress</u>	<u>IDate</u>	<u>ITime</u>	<u>Comments</u>	<u>StaffNo</u>	<u>SName</u>	<u>CarReg</u>
PK12	10-A, Model Town, Lahore	05-07-22	09:30	Roof repair needed	SG24	Saim Ghani	LM-5678
		15-08-22	11:00	In good order	AJ11	Ali Javed	PB-9876
		1-10-22	9:00	Bathroom tiles replacement	SG24	Saim Ghani	LW-3421
PK17	8-B, Gulberg, Karachi	02-06-22	10:45	Replace living room carpet	FA09	Farah Ahmed	LM-5678
		25-08-22	12:30	Good condition	SG24	Saim Ghani	PB-9876

- Find all candidate keys.
- Identify the best normal form that the above table satisfies and provide reasoning (1NF, 2NF, 3NF, or BCNF).
- Is this relation in BCNF? If not, show all dependencies that violate it.
- Is this relation in 3NF? If not, show all dependencies that violate it.
- Discuss potential issues related to updates on this table, including insertion, deletion, and modification anomalies.
- Describe and illustrate the process of normalizing the relation shown in above Table to 4NF.

Answer # 8(A)

1. Candidate Keys

The candidate key is **(PropertyNo, IDate)**. This combination uniquely identifies each inspection since a property is assessed only once per day.

Answer # 8(B)

1. Normalised Form

The normalised form that satisfies the given relation is a 1NF form.

2. Reason

The table satisfies **1NF** because all attributes have atomic values. It violates 2NF due to partial dependencies (e.g., PropertyNo $\rightarrow$ PAddress) and so on.

Answer # 8(C)

1. BCNF Check

NO, this relation does not satisfy BCNF conditions, thus it's not in BCNF form.

2. Dependencies

The dependencies that violate BCNF are as follows:

1. **StaffID $\rightarrow$ StaffName**
2. **VehicleID $\rightarrow$ VehicleModel**
3. **PropertyID $\rightarrow$ PropertyAddress**

Answer # 8(D)

1. 3NF Check

NO, this relation does not satisfy 3NF conditions, thus it's not in 3NF form.

2. Dependencies

The dependencies that violate 3NF are as follows:

1. **StaffID $\rightarrow$ StaffName**
2. **VehicleID $\rightarrow$ VehicleModel**

Answer # 8(E)

1. Update Anomalies

The update anomalies that can be formed in this table are as follows:

1. Insertion Anomaly

The insurability anomalies that can occur in the table are as follows:

1. *Cannot add a new property without an inspection record.*

2. Deletion Anomaly

The deletion anomalies that can occur in the table are as follows:

1. *Deleting all inspections for a property erases its address.*

3. Modification Anomaly

The Modification anomalies that can occur in the table are as follows:

1. *Changing a property's address requires updates across all its inspection records.*

Answer # 8(F)

3. Description/Illustration

The process of normalising the attributes shown in the table to produce a set of well-designed 4NF relations consists of several steps, which are as follows:

1. Step #1

1NF

Since the table satisfies 1NF, thus;

1. *INSPECTION (IDate, ITime, StaffID, StaffName, VehicleID, VehicleModel, PropertyID, PropertyAddress)*

2. Step #2

2NF

The second step consists of removing partial dependencies from the relation by creating a separate row for each entry.

1. *INSPECTION (PropertyID, IDate, ITime, StaffID, VehicleID)*
2. *PROPERTY (PropertyID, PropertyAddress)*
3. *STAFF (StaffID, StaffName)*
4. *VEHICLE (VehicleID, VehicleModel)*

3. Step #3

3NF

The third step involves removing transitive dependencies from the relation by creating a separate row for each entry.

After reviewing the relations created in step 2, we discovered the following:

1. *There are no transitive dependencies*

4. Step #4

BCNF

The fourth step consists of removing the overlapping of candidate keys from the relation by creating separate relations.

1. *Every determinant is a candidate key. All functional dependencies are enforced via primary keys.*
2. *There are no overlap on candidates keys.*

5. Step #5

4NF

The fifth step consists of removing multivalued dependencies from the relation.

1. INSPECTION (PropertyID, IDate, ITime)
 2. INSPECTION_STAFF (PropertyID, IDate, StaffID)
 3. INSPECTION_VEHICLE (PropertyID, IDate, VehicleID)
 4. PROPERTY (PropertyID, PropertyAddress)
 5. STAFF (StaffID, StaffName)
 6. VEHICLE (VehicleID, VehicleModel)
-